ABN SEAL COLLEGE
COOCHBEHAR
PANCHANAN BARMA UNIVERSITY
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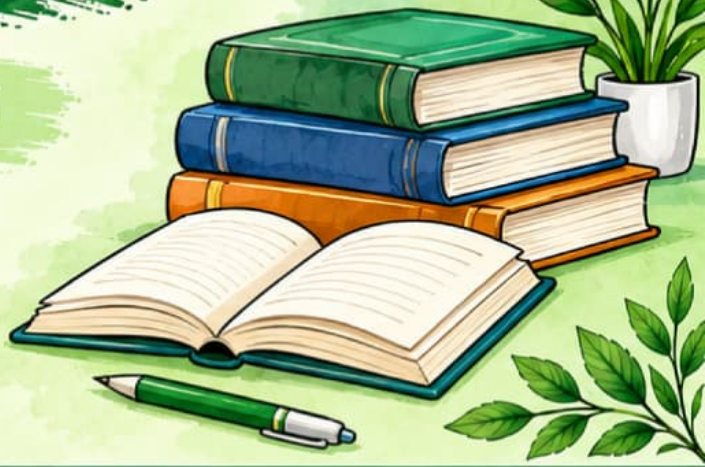


Think
Prepare
Excel

Some Important Questions



B.Sc. 2nd Semester | CBPBU



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Adaptations.

1) Hydrophyte ? types ? Adaptations. } ① Morphological ② Anatomical ③ Physiological + Ecological
2) Halophyte Adaptations.
3) Xerophyte Adaptations.
4) Significance of sunken stomata.
5) succulent ~~hydro~~ xerophytes v/s non-succulent ~~hydro~~ xerophytes.
6) Aerenchyma.
7) Pneumatophores.
8) Vivipary
9) water conservation mechanisms of Xerophytes.
10) Salt tolerance mechanisms of halophytes.

Dynamics & Succession.

1) what is succession ? Dessribe the different steps involved in the process of succession.
2) Discuss Hydrosere & Hydrach succession.
3) Discuss Xerosere & Xerarch succession.
4) Discuss briefly about primary succession.
5) Discuss briefly about secondary succession.
6) climax community. ②/④
7) primary succession v/s secondary succession.
8) types of succession —
   - primary, secondary succession
   - Autotrophic, Heterotrophic succession
   - Autogenic, Allogenic succession
   - progressive, Retrogressive succession.
   - Induced, cyclic succession.
9) Explain climax theory & its types.
10) Briefly describe the significance of ecological succession.
11) Classify ecological succession OTBO habitat, origin & Trophic nature.

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Ecology introduction

1) Define ecology.
2) Explain levels of organization of / ecological organization
3) Autoecology v/s synecology.
4) Biome v/s ecosphere.
5) Population v/s community.
6) Explain Biotic & Abiotic components of an ecosystem.
7) Define homeostasis & explain its ecological significance.
8) Explain +ve & -ve feedback mechanisms.
9) Discuss resilience & resistance in ecosystem.
10) Define BOD & explain its significance.
11) Define COD & " " ".
12) BOD v/s COD.

Biotic interaction

1) Trophic organization.
2) Trophic level.
3) Define autotrophy.
4) Define heterotrophy.
5) Define symbiosis.
6) explain +ve, -ve & neutral interaction among organisms. Describe different types of biotic interaction ⊕ in an ecosyste
7) what is epiphyte. ex.
8) what is lichen?
9) what is mycorrhiza?
10) Intraspecific v/s Interspecific competition.
11) What is Food chain?
12) " " Food web?
15) What are the different type of pyramid ⊕ in a population?
13) Food chain v/s Food web.
14) Discuss the role of decomposers in food chain.
16) upright v/s inverted pyramid.
17) Why pyramid of energy is always upright
18) Explain Lindman's 10% law.
19) What is Biomass?
20) Explain standing crop & standing state.

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Ecosystem

1) Define ecosystem & explain its components. (Biotic & Abiotic)
2) Input & output of ecosystem.
3) explain producer consumer & decomposer of ecosystem with example.
4) explain stratification in ecosystem.
5) Discuss types of ecosystem.
6) Describe the structure of pond ecosystem.
7) " " " " forest " .
8) " " " " Grassland " .
9) Explain ecosystem processes & its importance

Functional aspects of ecosystem

1) what are the major function of an ecosystem.
2) what is production & productivity.
3) production v/s productivity.
4) Types of ~~production~~ productivity in ecosystem
5) what is ecological efficiency.
6) what is age pyramid
7) what is models of energy flow & their types.
8) write a short note on ecological efficiency & their types.
9) Discuss ɤ shaped energy flow model in an ecosystem

Biogeo-chemical cycles

1) what do you mean by Biogeochemical cycle
2) Schematically represent N_2, P & C cycle

Population ecology

1) What is population growth? mention its types.
2) Ecological density & crude density.
3) Define survivorship curve. Describe different types of survivorship curve.
4) what is r & K selected species
5) r selected species v/s K selected species.
6) Ecological speciation. types.

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1) Explain Phytogeography, factor affecting it & Branches
2) What are the principles of Phytogeography.
3) Dispersal v/s migration.
4) Endemism definition, types.
5) Types of endemic plants based on cytogenetic endemism
6) write a note on - Theory of Tolerance
7) Define phytogeography. Describe the phytogeographical regions of India according to D. Chatterjee (1939, 1960).
8) Vegetation of Eastern Himalaya
9) Briefly describe the endemic flora of West Bengal.
10) Vegetation v/s Flora.
11) Continental Drift Hypothesis.

1) Define habitat & niche. Differentiate between them with suitable examples.
2) Define different types of Niche ⊕ in an ecosystem.
3) Niche width & Niche overlap - Definition with diagram.
4) Fundamental v/s Realized niche.
5) Analytical v/s synthetic niche.
6) Characters displacement.
7) Ecotone, Edge effect
8) Ecological Amplitude → types.
9) Keystone species.

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All important differences in Sem-2 (Covering entire syllabus)

1) Succulent xerophyte v/s Non succulent xerophyte.
2) primary succession v/s secondary succession.
3) Autoecology v/s Synecology
4) Biome v/s Biosphere
5) population v/s community
6) Resistance v/s Resilience
7) Trophic organization v/s Trophic level
8) Intraspecific competition v/s Interspecific competition
9) Food chain v/s Food web.
10) Standing crop v/s standing stock.
11) BOD v/s COD
12) Fundamental v/s ~~rel~~ realized niche
13) r-selected v/s k-selected species
14) production v/s productivity.
15) primary productivity v/s secondary productivity.
16) GPP v/s NPP

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